Lab Report - 06

Course Code : CSE 412

Course Title : Computer Graphics Lab

Experiment No : 06

Experiment Name : Implementation of Midpoint Circle drawing Algorithm

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Lab Report: 06

Implementation of Midpoint Circle drawing Algorithm

1. Title: Drawing a Circle Using Midpoint Circle Algorithm

Introduction:

In this project, a circle is drawn using the Midpoint Circle algorithm. The program calculates points and plots them to form a circle.

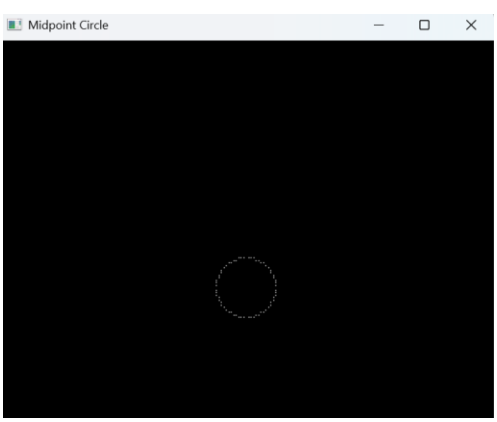
Contents:

- **Function Used:** `glBegin(),glVertex2i(),glColor3f()`
- **Algorithm Used:** Midpoint Circle Algorithm
- **Shape Used:** Circle

Graph:

```
Enter radius 12

Process returned 0 (0x0)   execution time : 28.780 s
Press any key to continue.
```



Draw X and Y axis and sketch a circle with a given radius.

Code (Drawing Portion Only):

```
glBegin(GL_POINTS);
while(x <= y)
{
    if(pk < 0)
    {
        x = x + 1;
        pk = pk + (2*x) + 1;
    }
    else
    {
        x = x + 1;
        y = y - 1;
        pk = pk + (2*x) + 1 - (2*y);
    }

    glVertex2i(x,y);
    glVertex2i(y,x);
    glVertex2i(-x,y);
    glVertex2i(-y,x);
    glVertex2i(-x,-y);
    glVertex2i(-y,-x);
    glVertex2i(x,-y);
    glVertex2i(y,-x);
}

glEnd();
```

Discussion:

The task was to draw a circle using the Midpoint Circle algorithm. The program calculates symmetric points and plots them to create the circle.

2. Title: Drawing an S Shape Using Midpoint Circle Algorithm

Introduction:

In this project, an S shape is created using the Midpoint Circle algorithm. Circle points are plotted to form the shape.

Contents:

- **Functions Used:** glBegin(), glVertex2i(), glColor3f()
- **Algorithm Used:** Midpoint Circle Algorithm
- **Shape Used:** Circle points forming an **S shape**

Graph:

Draw X and Y axis and sketch an **S shaped figure using circle arcs**.

Code (Drawing Portion Only):

```
glBegin(GL_POINTS);

while(x <= y)
{
    if(pk < 0)
    {
        x = x + 1;
        pk = pk + (2*x) + 1;
    }
    else
    {
        x = x + 1;
        y = y - 1;
        pk = pk + (2*x) + 1 - (2*y);
    }

    glVertex2i(6 + x , y);
    glVertex2i(6 - x , y);
    glVertex2i(6 + y , x);
    glVertex2i(6 - y , x);

    glVertex2i(-6 + x , -y);
    glVertex2i(-6 - x , -y);
    glVertex2i(-6 + y , -x);
    glVertex2i(-6 - y , -x);
}
glEnd();
```

Discussion:

The task was to draw an S shape. The Midpoint Circle algorithm was used to calculate circle points and plot them symmetrically to create the shape.

3. Title: Drawing a Circle Using OpenGL

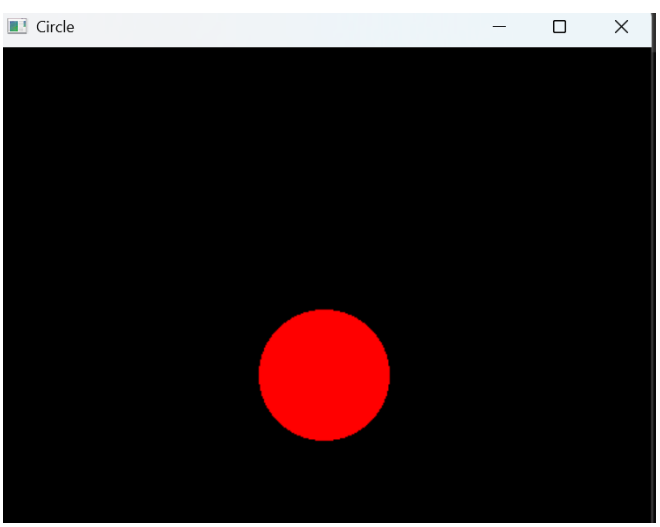
Introduction:

In this project, a circle is drawn using OpenGL. The circle is created by plotting points around a center.

Contents:

- **Functions Used:** `circle(),glBegin(),glVertex2f(),glColor3f()`
- **Shape Used:** Circle

Graph:



Draw X and Y axis and sketch a triangle using three points.

Code (Drawing Portion Only):

```
void circle(GLfloat rx,GLfloat ry,GLfloat cx,GLfloat cy)
{
    glBegin(GL_TRIANGLE_FAN);
    glVertex2f(cx,cy);

    for(int i=0;i<=6;i++)
    {
        float angle = 2 * 3.1416 * i/6;
        float x = rx * cos(angle);
        float y = ry * sin(angle);

        glVertex2f(x+cx,y+cy);
    }

    glEnd();
}
```

Discussion:

The task was to draw a circle. The program calculates points using angle and plots them around the center to form the circle.